Status	Finished
Started	Sunday, 20 September 2026, 1:42 PM
Completed	Monday, 21 September 2026, 2:23 PM
Duration	1 day
Marks	195.94/270.00
Grade	**7.26** out of 10.00 (**72.57**%)

Question 1

Incorrect

Mark 0.00 out of 10.00

Converting MIPS assembly instructions to machine language involves understanding the instruction formats

Instruction: `sll $t0, $t1, 8`

Machine instruction is: 0b 00000000000010000100101000000000(

Question 2

Incorrect

Mark 0.00 out of 10.00

Converting MIPS assembly instructions to machine language involves understanding the instruction formats

Instruction: `and $t0, $t1, $t2`

Machine instruction is: 0b 00000001000010010101000000100100

Question 3

Incorrect

Mark 0.00 out of 10.00

Converting MIPS assembly instructions to machine language involves understanding the instruction formats

Instruction: `nor $t8, $t5, $t9`

Machine instruction is: 0b 00000010000011011000100000100111

Question 4

Incorrect

Mark 0.00 out of 10.00

Converting MIPS assembly instructions to machine language involves understanding the instruction formats

Instruction: `srlv $t7, $t9, $t8`

Given that `$t8` stores `12`

Machine instruction is: 0b 00000001111100011000001100000110

Question 5

Incorrect

Mark 0.00 out of 10.00

Converting MIPS assembly instructions to machine language involves understanding the instruction formats

Instruction: `addu $12, $t9, $t6`

Machine instruction is: 0b 00000001100100010111000000100001

Question 6

Correct

Mark 10.00 out of 10.00

Let's execute the instruction `add $t7, $t3, $t4`. Given the values `$t3 = -8` and `$t4 = 1`.
The execution is performed in a **4-bit architecture**.

Demonstrate the data for the machine by

$t3 = 0b 1000

$t4 = 0b 0001

$t7 = 0b 1001

Carry flag = 0

Borrow flag = 0

Overflow flag = 0

Underflow flag = 0

Question 7

Partially correct

Mark 8.57 out of 10.00

Let's execute the instruction `add $t7, $t3, $t4`. Given the values `$t3 = -8` and `$t4 = 1`.
The execution is performed in a 32-bit architecture.

Demonstrate the data for the machine by

$t3 = 0x fffffff8 ✓

$t4 = 0x 00000001 ✓

$t7 = 0x fffffff7 ✗

Carry flag = 0 ✓

Borrow flag = 0 ✓

Overflow flag = 0 ✓

Underflow flag = 0 ✓

Question 8

Partially correct

Mark 46.34 out of 50.00

Let's execute the instruction `addu $t7, $t3, $t4`. Given the values `$t3 = -8` and `$t4 = 1`.
The execution is performed in a 32-bit architecture.

Demonstrate the data for the machine by

$t3 = 0x fffffff8 ✓

$t4 = 0x 00000001 ✓

$t7 = 0x fffffff7 ✗

$t7 = 0d -7 ✗

$t7 = 0b 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓
1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓
1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 1 ✓ 0 ✓ 0 ✓ 1 ✓

Carry flag = 0 ✓

Borrow flag = 0 ✓

Overflow flag = 0 ✓

Underflow flag = 0 ✓

The machine instruction encoding for `addu $t7, $t3, $t4`, it is 0x 016a7821 ✗

Question 9

Correct

Mark 10.00 out of 10.00

Let's execute the instruction `add $t8, $t3, $t6`. Given the values `$t3 = 8` and `$t6 = 1`. And the execution is performed in a MIPS 32-bit architecture.

Demonstrate the data for the machine by

$t3 = 00000000000000000000000000001000

$t6 = 00000000000000000000000000000001

$t8 = 00000000000000000000000000001001

Question 10

Partially correct

Mark 48.78 out of 50.00

Let's execute the instruction `addu $t6, $t3, $t2`. Given the values `$t3 = -8` and `$t2 = -1`.
The execution is performed in a 32-bit architecture.

Demonstrate the data for the machine by

$t3 = 0x fffffff8

$t2 = 0x ffffffff

$t6 = 0x fffffff7

$t6 = 0d 7

$t6 = 0b 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 1 1 1

Carry flag = 1

Borrow flag = 0

Overflow flag = 0

Underflow flag = 0

The machine instruction encoding for `addu $t6, $t3, $t2`, it is 0x 016a7021

Question 11

Partially correct

Mark 42.25 out of 50.00

Let's execute the instruction `subu $t0, $t1, $t2` with the values `$t1 = 5` and `$t2 = 10` on a 32-bit MIPS architecture.

Demonstrate the data for the machine by

$t1 = 00000000000000000000000000000101

$t2 = 00000000000000000000000000001010

$t0 = 11111111111111111111111111111011

Evidence:

Carry: No

Overflow: No

Underflow: No

Zero: No

Instruction: `subu $t0, $t1, $t2`

= 0b 0 0 0 0 0 0 0 1

0 0 0 0 1 0 0 1

0 1 0 1 0 0 0 0

0 0 1 0 0 0 1 1

= 0x 01096023

Question 12

Correct

Mark 25.00 out of 25.00

In an MIPS architecture, let's use the instruction: `nor $t0, $t1, $t2`.

Given:

- `$t1 = 0x0000AAAA`
- `$t2 = 0x00005555`

In the binary code (4 bits (binary per block)

$t1 =0b 0000 0000 0000 0000 1010 1010 1010 1010

$t2 =0b 0000 0000 0000 0000 0101 0101 0101 0101

$t0 =0b 1111 1111 1111 1111 0000 0000 0000 0000

$t0 is 0x ffff0000

Question 13

Partially correct

Mark 5.00 out of 15.00

In the MIPS architecture, let's execute the instruction `sll $t0, $t1, 5` where `$t1 = 32`.

Demonstrate the data for the machine by

$t1 = 00000000000000000000000000000101

$t0 = 00000000000000000000000001010000

Result in decimal number format is 1024